Data Science Project Planning: North Quay Gifts

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1. Identification and prioritization of opportunities

- Chosen Opportunity: Prediction of Net Customer Value (adjusted for cancellation risk and revenue concentration impact).
- Justification: This opportunity is prioritized because it recognizes that high gross billing does not equal real value if it entails high cancellation rates. Instead of treating all buyers equally, this approach identifies the critical percentage of customers that sustain the revenue (Pareto Principle) and projects their net future spending. The general weekly sales forecast (Alternative 3) is postponed, as predicting the individual behavior of high-impact customers is more actionable for retention and business profitability.

2. Business Approach

- Problem and context: The company relies on wholesale and retail customers who show heterogeneous behaviors. Currently, global historical revenues may be inflated by orders that are subsequently canceled (invoices starting with 'C'). The lack of differentiation prevents knowing which customers are truly valuable for the future.
- Area and business decision: Finance and Commercial Management. It will support the decision to prioritize commercial relationships, allocate loyalty budgets, and mitigate the logistical risk of customers with high return rates.
- Business objective: Estimate the future net spending of each customer to prioritize commercial efforts and avoid relying on seemingly large accounts that have high cancellation rates.
- Expected benefit: Resource optimization, reduction of reverse logistics costs due to returns, and financial planning based on probable real revenues rather than just gross historical figures.
- Business success criterion: Increase in commercial return on investment (ROI) and greater accuracy in the cash flow projection for the next quarter.
- Information to measure it: Internal information on product costs, profit margins, and shipping/reverse logistics costs would be required to calculate absolute financial profitability.

3. Investigation of real references

Element	Case 1 (Customer Lifetime Value)	Case 2 (Return Prediction)
Organization and application	Olist & Hybrid Retail: CLV prediction for budget optimization.	Fashion E-commerce (Germany): Real-time product return prediction.

Verifiable source	ResearchGate (2025). Forecasting Customer Lifetime Value: A Data-Driven Approach.	ResearchGate (2015). Predicting Product Returns in E-Commerce: The Contribution of Mahalanobis Feature Extraction.
Business problem	Need to identify long-term high-value customers to properly allocate marketing spend amid future behavior uncertainty.	Need to avoid high logistical costs associated with product returns by identifying transactions with a high probability of return.
Data and analytical task	Use of transactional, behavioral, and demographic data. Task: Regression (CLV prediction and churn probability).	Use of a dataset of over 1 million purchases. Task: Classification (probability of a shopping cart being returned).
Method or algorithm	Hybrid Deep Learning model combined with probabilistic models (BG/NBD) and Random Forest.	Mahalanobis feature extraction and Machine Learning algorithms (sparse matrix encoding).
Results and impact	The model reduced prediction error and improved customer ranking quality, translating into reported gains in Return on Marketing Investment (ROMI).	Identified indicators at the customer and cart level that allow predicting returns, enabling interventions prior to the transaction.
Relevance and limits	Highly relevant due to the focus on retail and repeat sales. Limit: North Quay lacks demographic and marketing data.	Demonstrates that a customer's history predicts future returns. Limit: North Quay is not a fashion retailer and lacks rich product attributes.

- Transfer analysis: The references support that transactional histories (recency, frequency, and spending) and past return behavior are robust predictors of a customer's future value. However, for North Quay Gifts, an increase in pure profitability (ROMI) cannot be assumed, as we lack the complete context of margins and acquisition costs; our prediction will be limited to sales volume net of cancellations.

4. Situation assessment

- Starting data: Transactional dataset with dates, invoices, customer ID, country, product codes, quantity, price, and invoices with a 'C' prefix for cancellations.
- Unit of analysis: The customer (Customer ID).

- Relevant variables: Recency of the last purchase, historical order Frequency, Monetary Value (price x quantity), and historical cancellation rate per customer (derived from 'C' invoices).
- Quality and known limits: There are customers without IDs that cannot be tracked, negative quantities, and no explicit churn or logistics cost labels.
- Additional information: Shipping costs, product costs, profit margins, and the explicit reason for cancellations.
- Assumptions and restrictions: It is assumed that the analyzed period (2009-2011) reflects the normal seasonality of the business and that a customer inactive for X months is not necessarily lost permanently.

Risk	Possible consequence	Response or contingency
1: Missing IDs and data noise	Cannot calculate historical value or cancellation risk for anonymous sales, reducing usable data volume.	Filter out records without ID for the modeling phase and group these transactions as an independent "baseline demand".
2: Mix of Wholesalers and Retailers	Atypical wholesale purchase patterns could bias the general model, affecting predictions for smaller retail customers.	Create an initial segmentation (Pareto) and train separate models or add a categorical customer profile variable.
3: Absence of margins and costs	Management could interpret the projected spending metric as real profitability, leading to poor financial decisions.	Communicate transparently and continuously that the result is a projection of gross revenues adjusted for cancellations, not profit.

5. Data Science Objective

- Data Science Objective: Estimate the expected spending generated by each customer in the next 3 to 6 months, adjusting this figure by their individual probability of cancellation.
- Preliminary analytical task: Regression (for the monetary estimate) combined with Classification (probability of return).
- Unit and horizon of analysis: Customers with known IDs; using the first year of data as training and projecting a 3 to 6-month horizon.
- Expected result: A dashboard or database assigning each customer a "Projected Net Value" score and their "Cancellation Risk Level".
- Results evaluation: Purchases from the last months of the historical dataset will be reserved to check if the model's predictions align with what was actually observed later.
- Technical success criterion: Reduce the Mean Absolute Error (MAE) of predicted future sales by 20% compared to a simple prediction based solely on the customer's historical average.

- Relationship with the business: Will provide the Commercial team with a prioritized ranking to focus their loyalty resources on the 20% of customers (Pareto) that ensure the highest net revenue free of returns.

6. Project Plan (CRISP-DM)

CRISP-DM Phase	Planned activities	Deliverable or milestone	Responsible
Business Understanding	Meetings with Finance and Commercial to define business rules on returns and revenues.	Scope document and success metrics.	Full Team
Data Understanding	Descriptive analysis of Online Retail II, quantifying customers without ID and 'C' invoices.	Data quality report.	Member 1
Data Preparation	Cleaning duplicates, calculating RFM metrics, and assigning return rates per customer.	Structured dataset by Customer.	Member 2
Modeling	Training regression and classification algorithms. Identification of key customers (Pareto).	Preliminary trained models.	Member 3
Evaluation	Comparison of model predictions against reserved history (final months of 2011).	Model validation report.	Full Team
Deployment / Adoption	Presentation of results and generation of a list of customers to prioritize or contact.	Executive presentation and scoring list.	Full Team

Six-month Timeline

Phase	M1	M2	M3	M4	M5	M6
Business Understanding	X					
Data Understanding	X	X				
Data Preparation		X	X			

Modeling			X	X		
Evaluation					X	
Deployment / Adoption						X

7. Executive Recommendation

We recommend that management initiate the project by focusing on the Prediction of Net Customer Value, an initiative that combines the business's revenue concentration with cancellation prediction. Addressing this problem ensures focusing limited resources on the small group of buyers who truly sustain the business model, while discounting high-logistics-risk profiles. Evidence in the retail sector has shown that modeling purchase and return history improves budget allocation and reduces costs associated with reverse logistics. Although our current dataset has limitations, such as the absence of costs or reasons for churn, a 6-month approach will deliver an objective customer ranking verifiable against historical data. This is the most defensible opportunity to ensure high-quality revenues before scaling to general forecasting systems.

References used by the team

- ResearchGate (2025). Forecasting Customer Lifetime Value: A Data-Driven Approach to Optimizing Marketing Budget Allocation. Retrieved from https://www.researchgate.net/publication/396713191_Forecasting_Customer_Lifetime_Value_A_Data-Driven_Approach_to_Optimizing_Marketing_Budget_Allocation
- ResearchGate (2015). Predicting Product Returns in E-Commerce: The Contribution of Mahalanobis Feature Extraction. Retrieved from https://www.researchgate.net/publication/283270887_Predicting_Product_Returns_in_E-Commerce_The_Contribution_of_Mahalanobis_Feature_Extraction